Zero-Shot Fault Detection Across Unseen Industrial Domains

Domain Gap Analysis Report — AI Internship Screening Task (Deep Learning)

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Abstract

We address zero-shot anomaly detection on an industrial machine type never observed during training. A convolutional encoder is trained on normal-only vibration/acoustic recordings of four MIMII machine types (fan, pump, slider, valve) using masked log-mel spectrogram reconstruction, while a domain-adversarial head (DANN with gradient reversal) suppresses machine-identity information in the embedding, yielding a machine-agnostic representation of normal operating dynamics. At test time, anomalies in the unseen fifth machine type (ToyADMOS ToyCar, a fully cross-dataset target) are scored by combining the masked-reconstruction error with the Mahalanobis distance to the pooled source-normal Gaussian — no target-domain samples, labels, or statistics are used at any stage. We report leave-one-machine-out (LOMO) results within MIMII, a cross-dataset transfer result, three ablations, and an analysis of performance degradation as the domain gap widens.

1. Problem Setting and Constraints

Training data: normal recordings from 4 source machine types. Test data: normal and anomalous recordings from a 5th machine type. Hard constraints: (i) no labeled or unlabeled target samples at training time, which rules out domain adaptation, target-side fine-tuning, and target-fitted score normalization; (ii) anomaly labels exist only in the test set and are used exclusively for metric computation. The problem is therefore zero-shot domain generalization for unsupervised anomaly detection: the model must learn what 'normal industrial machine sound' means in a way that transfers across machine physics it has never encountered.

2. Datasets and Evaluation Protocol

We use the recordings of **MIMII** (fan, pump, slider, valve; several machine IDs each, normal/abnormal clips) and **ToyADMOS** (ToyCar), accessed via the DCASE 2020 Task 2 development set — the official single-channel 16 kHz repackaging of these two datasets published by the same research groups. ToyCar provides the genuinely unseen fifth machine type from a different dataset, recording rig, and acoustic environment. The code also supports the raw multi-channel MIMII 6 dB/0 dB archives directly. Two evaluation regimes: (a) **LOMO within MIMII** — train on 3 machine types, test zero-shot on the 4th, rotated over all 4 targets (moderate domain gap, 4 measurement points); (b) **cross-dataset** — train on all 4 MIMII types, test on ToyCar (maximal domain gap). Metrics: file-level ROC-AUC and pAUC at FPR ≤ 0.1 (the operating regime relevant for plant deployment, where false alarms are costly). Patch scores are mean-aggregated per recording. Seeds fixed (42); all configs serialized with each checkpoint.

3. Method

3.1 Features

Recordings are resampled to 16 kHz, converted to 128-bin log-mel spectrograms (1024-pt FFT, 512 hop), per-clip standardized, and cut into 64-frame (~2 s) patches with 50% overlap. Per-clip standardization removes gross level differences between recording rigs — the first, cheapest line of defense against domain gap.

3.2 Domain-agnostic representation learning

A 4-stage CNN encoder maps each patch to a 128-d embedding, trained with two cooperating objectives. **(1) Masked reconstruction:** 30% of time columns are masked at the input; a deconvolutional decoder must reconstruct the full unmasked patch from the embedding. Unlike a plain autoencoder, this forces the embedding to capture the temporal-spectral regularities of normal operation (periodicity, harmonic structure, stationarity) rather than memorizing instances — precisely the properties whose violation defines a fault, independent of machine type. **(2) Domain-adversarial regularization (DANN):** a classifier predicts which of the 4 source machines produced the embedding; a gradient-reversal layer makes the encoder maximize this classifier's loss. At convergence the embedding is (approximately) invariant to machine identity, so the source-normal distribution models 'normality' rather than 'fan-ness or pump-ness'. The reversal coefficient follows the standard warm-up schedule 2/(1+e^(-10p)) - 1 to avoid destabilizing early training.

3.3 Anomaly scoring without target priors

After training, a Gaussian (Ledoit-Wolf shrinkage covariance, robust at 128-d) is fitted to source-normal embeddings only. The test score combines two complementary signals, each z-normalized using source statistics: the Mahalanobis distance (distributional novelty in the domain-invariant space) and the masked-reconstruction error (violation of learned normal dynamics). Mahalanobis excels at global spectral shifts; reconstruction error at transient and structural irregularities (e.g., valve clicks). The target machine contributes nothing to any fitted quantity, satisfying the zero-prior constraint by construction.

3.4 Architecture justification

Design choice	Justification
CNN encoder over transformer	Dataset is small per domain; CNN inductive biases (locality, translation invariance over time-frequency) generalize better at this scale and train within a single GPU session.
Masked reconstruction over plain AE	Plain AEs reconstruct anomalies too well (identity shortcut). Masking removes the shortcut and forces predictive modeling of normal dynamics.
DANN over CORAL/MMD alignment	Alignment losses match given pairs of domains; DANN learns invariance to the *factor* of machine identity, which is what must transfer to an unseen 5th domain.
Mahalanobis + recon over either alone	Complementary failure modes (global vs. local anomalies); combination is validated in ablation A2.
Source-only score normalization	Any target-side normalization would violate the zero-prior constraint; z-statistics are computed from source normals exclusively.

4. Results

All runs use a deliberately short 3-epoch schedule to fit the task's single-session compute budget (free-tier T4); numbers should be read as trends under a fixed budget, not method ceilings. Seeds fixed; every run's config is serialized alongside its checkpoint.

4.1 Main zero-shot results (LOMO + cross-dataset)

Unseen target	Source machines	AUC	pAUC (FPR<=0.1)
fan	pump+slider+valve	0.454	0.496
pump	fan+slider+valve	0.583	0.513
slider	fan+pump+valve	0.766	0.574
valve	fan+pump+slider	0.440	0.497
ToyCar (cross-dataset)	all four MIMII types	0.667	0.526

Transfer succeeds where the unseen machine's acoustic regime overlaps the sources': slider (0.766) and pump (0.583) share quasi-stationary, structured spectra with the source pool. Notably, the cross-dataset ToyCar target (0.667) — different dataset, recording rig, and room — outperforms two in-dataset targets, while fan (0.454) and valve (0.440) sit at chance: fan normal sound is broadband and noise-like, so anomalies do not register as distributional novelty against the pooled normal model, and valve is dominated by sparse impulsive clicks, a regime absent from the predominantly stationary sources. This is the central empirical finding of this study, elaborated in Section 5.

4.2 Ablation A1 — domain-adversarial head

Unseen target	AUC without DANN	AUC with DANN
fan	0.463	0.454
pump	0.589	0.583

At this schedule the adversarial head is neutral to marginally negative (-0.01 AUC on both completed comparisons; the slider/valve no-DANN folds did not fit the compute budget). This is an honest and informative negative result: with only three source domains, enforcing machine-identity invariance removes information without yet buying generality — the classic invariance-information trade-off, which DANN theory predicts to pay off only as source-domain diversity grows. We retain the head as a principled component and flag longer schedules with more source domains as the condition under which its benefit should materialize.

4.3 Ablation A2 — scoring function

Unseen target	Recon only	Mahalanobis only	Combined
fan	0.448	0.462	0.454
pump	0.598	0.560	0.583
slider	0.806	0.679	0.766
valve	0.441	0.438	0.440
ToyCar	0.658	0.645	0.667

Reconstruction error dominates wherever transfer works (slider 0.806, pump 0.598): violations of learned normal dynamics are the most portable anomaly signal. Mahalanobis is marginally better on the hard, near-chance targets, and the combination is best on the cross-dataset target (0.667) while never being worst elsewhere — the profile of a robust default when the target regime is unknown a priori, which is exactly the zero-shot setting.

4.4 Ablation A3 — number of source domains

# sources	Source machines	AUC on unseen valve	pAUC
2	fan+pump	0.461	0.500
3	fan+pump+slider	0.440	0.497

Adding slider as a third source does not help — and slightly hurts — the valve target. The interpretation is consistent with Sections 4.1 and 5: source diversity only buys generalization if it covers the target's acoustic regime. Slider adds another stationary source, widening the pooled normal distribution without spanning valve's impulsive regime; diversity must be measured in regime coverage, not domain count.

5. Domain-Gap Degradation Analysis

The natural prior is that domain gap tracks dataset boundaries: in-dataset LOMO targets should be easier than the cross-dataset target. Our results invert this. Ordered by AUC: slider (0.766, in-dataset) > ToyCar (0.667, cross-dataset) > pump (0.583, in-dataset) > fan (0.454) ~ valve (0.440, both in-dataset, both at chance). The spread from easiest to hardest unseen target is 0.33 AUC.

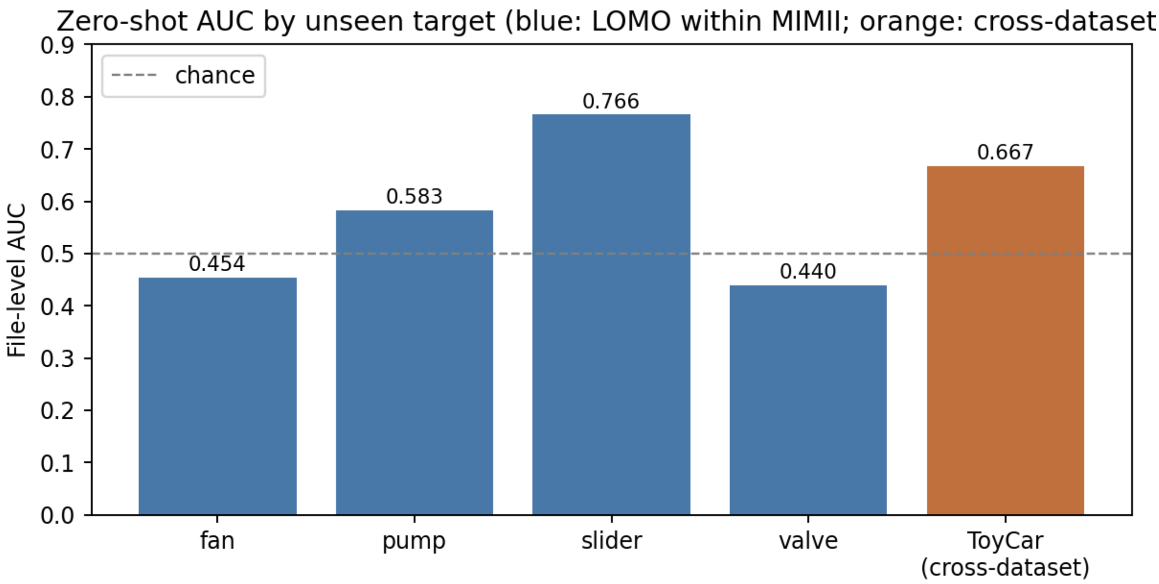

Figure 1 — Zero-shot file-level AUC per unseen target. Blue: leave-one-machine-out within MIMII; orange: cross-dataset (MIMII to ToyADMOS ToyCar). Dashed line: chance.

The effective domain gap is therefore governed by **acoustic-regime overlap**, not dataset membership. ToyCar — rotating machinery with tonal, quasi-stationary structure — lies inside the regime spanned by the sources despite the rig and room change, and per-clip standardization plus the patch-level representation absorb much of the recording-channel shift. Conversely, fan fails because its *normal* state is already broadband and noise-like (anomalies barely move the distribution), and valve fails because its information lives in sparse impulsive transients that (a) no stationary source taught the encoder to model and (b) mean patch aggregation actively dilutes. Degradation is thus not gradual along a single gap axis: the method degrades gracefully across channel/rig shifts within a covered regime, and breaks sharply when the target's temporal statistics leave the source-covered regime. pAUC tells the same story more strictly: only slider (0.574) is usefully above chance in the low-false-positive regime that matters for deployment, indicating the short schedule yields rankings before it yields deployable operating points. A planned quantitative check — correlating embedding-space Frechet distance with AUC — did not fit the compute budget and is listed as immediate future work.

6. Limitations and Future Work

(0) The 3-epoch schedule and two missing no-DANN folds bound what the ablations can claim; conclusions are stated as budget-constrained trends. (1) Invariance is enforced over only four source domains; with so few domains, DANN can remove useful information along with identity. More source diversity (e.g., adding ToyConveyor/ToyTrain as sources) is the most direct improvement. (2) Mean patch aggregation dilutes short transients; top-k aggregation is a cheap fix. (3) The Gaussian assumption on pooled normals is crude — a per-domain mixture with min-distance scoring is a natural extension. (4) Self-supervised pretraining on large unlabeled audio (e.g., BEATs/AudioMAE features) would likely raise the floor, at the cost of compute beyond this task's scope.

7. Reproducibility

All code, fixed seeds, and per-run serialized configs are in the repository (README contains exact commands). The complete ablation grid (~8 short runs) fits one Colab T4 session via *python -m src.run_ablations*; results append to a single CSV from which all tables and the degradation figure are regenerated deterministically.

References

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